The Journal of Open Source Software

BattMo - Battery Modelling Toolbox

Xavier Raynaud [1¶], **Halvor Møll Nilsen** [1], **August Johansson** [1], **Eibar Flores** [2], **Lorena Hendrix** [2], **Francesca Watson** [1], **Sridevi Krishnamurthi** [2], **Olav Møyner** [1], **and Simon Clark** [2]

1 SINTEF Digital, Dept. of Mathematics and Cybernetics, Oslo, Norway **2** SINTEF Industry, Dept. of Sustainable Energy Technology, Trondheim, Norway ¶ Corresponding author

DOI: 10.21105/joss.11114

Software

- Review
- Repository
- Archive

Editor: Mojtaba Barzegari

Reviewers:

- @ElektrikAkar
- @yuefan98

Submitted: 10 December 2025
Published: 23 October 2026

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

This paper presents the Battery Modelling Toolbox (BattMo), a flexible finite volume continuum modelling framework in MATLAB® (The MathWorks Inc., 2025) for simulating the performance of electrochemical cells. BattMo can be used to quickly set up and solve models for a variety of battery chemistries, even considering 3D designs such as cylindrical and prismatic cells.

The simulation input parameters, including the material parameters and geometric descriptions, are specified through JSON schemas. In this respect, we follow the guidelines of the Battery Interface Ontology (BattINFO) to support semantic interoperability in accordance with the FAIR principles (Wilkinson et al., 2016).

The Doyle-Fuller-Newman (DFN) (Doyle et al., 1993) approach is used as a base model. We include fully coupled thermal simulations. It is possible to include degradation mechanisms such as solid electrolyte interphase (SEI) layer growth and composite materials such as a mixture of silicon and graphite.

The models are set up in a hierarchical way, for clarity and modularity. Each model corresponds to a computational graph, which introduces a set of variables (the nodes) and functional relationships (the edges). This design enables the flexibility to modify existing models and design new ones.

The solver in BattMo uses automatic differentiation and supports adjoint computation. We can therefore compute the derivative of objective functions with respect to all parameters efficiently. Gradient-based optimization routines can be used to calibrate parameters from experimental data.

Statement of need

New high-performance electrochemical systems such as Li-ion and post-Li-ion batteries are essential to achieve the goals of the electric energy transition. Developing rigorous digital workflows can help industrial and research institutions reduce the need for physical prototyping and derive greater insight and knowledge from their data.

BattMo extends this effort by supporting fully 3D geometry and the possibility to easily modify the underlying equations. We provide a library of standard parameterized battery geometries. Design optimization can also be done on the geometry, which is an essential part of the cell design. BattMo is useful for experienced battery designers wanting to optimize and virtually test different designs, research software developers integrating simulation tools in electrochemical system workflows, and beginners wanting to understand fundamental processes.

A challenge for physics-based models of batteries and other electrochemical systems is the

difficulty of calibrating the parameters. With an adjoint-based approach, we can effectively calibrate the models from experiments in a reasonable computational time.

State of the field

Recently, a variety of open-source battery modelling codes have been released including PyBaMM (Sulzer et al., 2021), cideMOD (Ciria Aylagas et al., 2022), LIONSIMBA (Torchio et al., 2016), Dandeliion (Korotkin et al., 2021) and PETLion (Berliner et al., 2021), among others. These open-source frameworks help the community reduce the cost of model development and help ensure the validity and the reproducibility of findings.

PyBaMM is, to our knowledge, the most popular option. Its electrochemical models are primarily P2D, with so-called 2+1D model support for improved modelling of current collectors and thermal properties. In 2025, support for 3D thermal FEM was added, and its 3D functionality is currently expanding. For calibration and optimization, the developers recommend PyBOP (Planden et al., 2025), an external package tightly coupled to PyBaMM. PyBaMM's extensive portfolio of degradation models is a particular strength. Based on FEniCS (Alnæs et al., 2015), cideMOD supports simulations in 3D by design, although its lack of automatic differentiation makes handling complex nonlinear systems and performing design optimization more challenging. COMSOL (COMSOL AB, 2025) is a commercial, closed-source alternative. Unlike the battery-specific tools discussed above, it is a general-purpose multi-physics platform that enables coupling of a wide range of physical phenomena.

Within this portfolio of tools, BattMo's foundation is designed for coupled electrochemical-thermal simulations in 3D, as well as calibration and optimization. Its current portfolio of degradation models is more limited than PyBaMM's, with support for SEI growth through the Safari (Safari et al., 2009) and Bolay (Bolay et al., 2022) models, as well as lithium plating (Hein et al., 2020).

Software design

BattMo builds on the MATLAB Reservoir Simulation Toolbox (MRST) (Lie, 2019) which provides the foundation for meshing intricate geometries, solving large nonlinear systems of equations, visualizing the results, etc., in addition to fundamental robust, low-order finite volume methods and fully implicit time discretization. Although developed in MATLAB®, we try to provide Octave (Eaton et al., 2025) compatibility. Neither BattMo nor MRST rely on extra MATLAB® packages; the basic license is sufficient. We do recommend using AMG preconditioners from the open-source AMGCL library (Demidov, 2020) for fast, multi-threaded solution of linear systems.

Functionality overview

The main features of BattMo are summarized in the following list:

- JSON-based input with schema
- Library of parameterized battery formats
- Flexible graph-based model design
- Fully coupled electrochemical-thermal models
- 3D visualization
- Parameter calibration
- Design optimization
- Support for standard protocols such as CC, CV, CCCV, and time series
- SEI layer growth model
- Composite material model
- Silicon swelling model

- Material database for NMC, LCO, LFP, LNMO, SiGr, electrolytes, etc.
- Alkaline membrane electrolyser model
- Proton ceramic membrane model

Battery format library

We support coin cells, jelly roll cells, and multi-pouch cells with different tab layouts, see Figure 1. The geometries are parameterized and can be modified using a simple set of parameters. 1D and 2D grids for P2D and P3D models can also be generated.

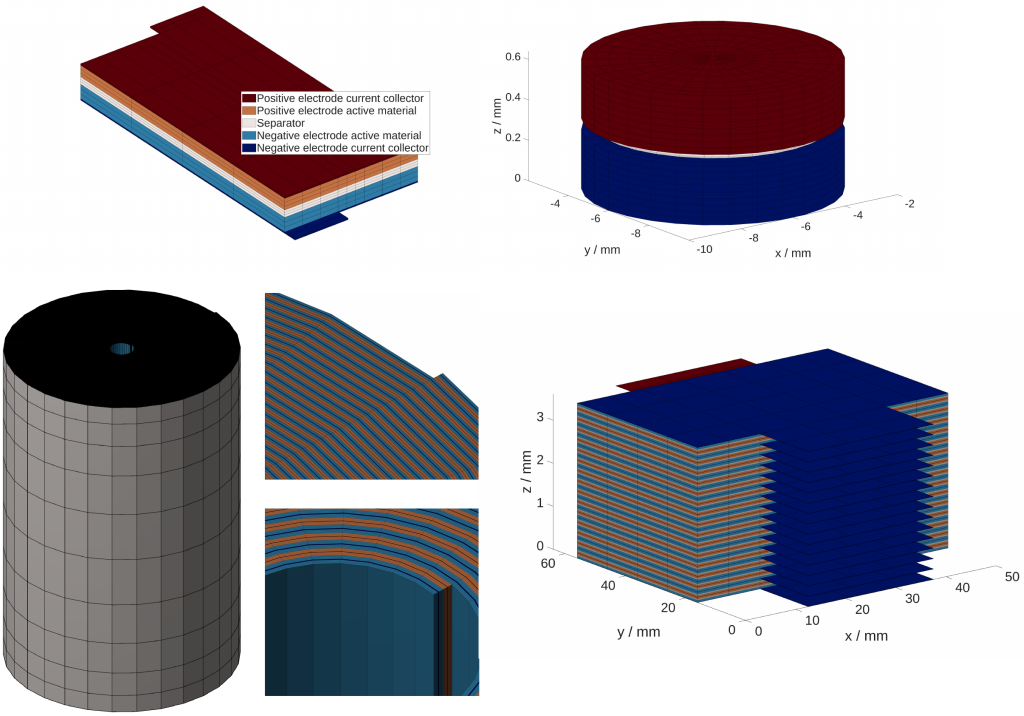


Figure 1: A selection of the parameterized battery geometries available. Clockwise from top left are: a single-layer pouch cell, CR 2016 coin cell, 30-layer pouch cell, and jelly roll cylindrical cell.

Graph-based model development

BattMo has the ambition to support a variety of electrochemical systems. The complexity of such models increases rapidly as models are extended or coupled. To manage this, BattMo introduces a computational graph-based model design. Each model is defined as a graph whose nodes represent variables and whose directed edges represent their functional relationships – see Figure 2. A model is ready for simulation when the graph's roots are the governing variables and its leaves are the governing equations. Interactive tools are available to explore these graphs.

Model hierarchy is an essential part of the framework. Coupling two models is done by creating a new coupling model that contains both as sub-models. Their graphs become sub-graphs, and new edges are added to represent coupling mechanisms, allowing most sub-models to remain unchanged. For more details, see the documentation.

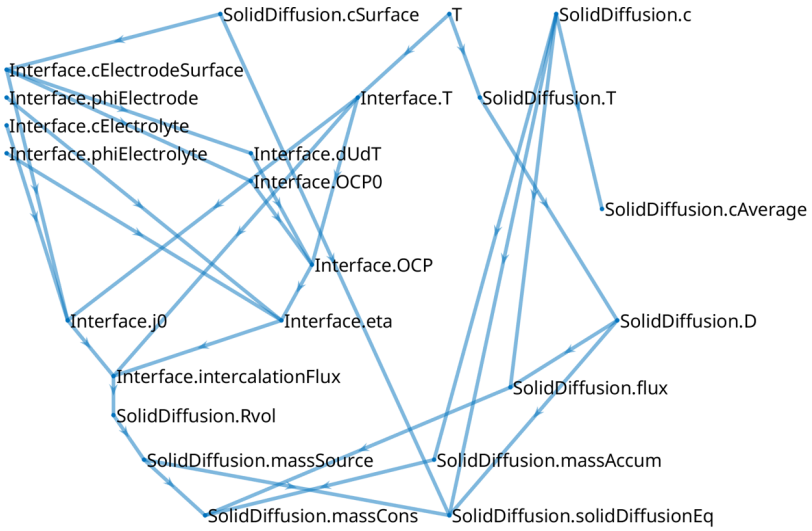


Figure 2: The computational graph of an active material model.

Examples

Numerous documented and tested examples are provided with the code and demonstrate the features listed above. We seek to be cross-compatible, allowing for any chemistry and material combination with any battery format. A selection of the examples is documented at https://battmo.org/BattMo/. The complete list of examples is available at https://github.com/BattMoTeam/BattMo/tree/main/Examples.

BattMo family

The following software packages are included in the BattMo family:

Software	Description
BattMo	MATLAB® version presented in this publication
BattMo.jl	Julia version
PyBattMo	Python wrapper around BattMo.jl
BattMoApp	Web application built on top of BattMo.jl

Research impact statement

By combining a modular modelling framework with documented JSON-based inputs, tutorials, and interfaces such as the Julia bridge, BattMo provides reproducible battery cell development workflows for research and education. BattMo has been used in several commercial and non-commercial projects. Publications include Clark et al. (2026) and Schmitt et al. (2026). It has been important for the success of several EU projects, primarily HYDRA (875527) and BATMAX (101104013), where it has been used to model novel LNMO cells, as well as commercial LFP and NMC cells in both 1D and 3D.

AI usage disclosure

Development of BattMo started before generative AI coding assistants became available. Therefore only very minor use of AI tools (ChatGPT from OpenAI and Microsoft Copilot) was

made in the development of this software. AI tools have been used to assist with phrasing and conciseness when writing this manuscript. In all instances, the authors have reviewed the suggestions and edited them if necessary.

Acknowledgements

BattMo has mainly been developed and used in projects funded by the European Union. We acknowledge contributions from the European Union, Grant agreements 101069765 (IntelLiGent), 875527 (HYDRA), 957189 (BIG-MAP), 101104013 (BATMAX) and 101103997 (DigiBatt).

References

Alnæs, M., Blechta, J., Hake, J., Johansson, A., Kehlet, B., Logg, A., Richardson, C., Ring, J., Rognes, M. E., & Wells, G. N. (2015). The FEniCS Project Version 1.5. *Archive of Numerical Software*, *Vol 3*. https://doi.org/10.11588/ANS.2015.100.20553

Berliner, M. D., Cogswell, D. A., Bazant, M. Z., & Braatz, R. D. (2021). Methods—PETLION: Open-Source Software for Millisecond-Scale Porous Electrode Theory-Based Lithium-Ion Battery Simulations. *Journal of The Electrochemical Society*, *168*(9), 090504. https://doi.org/10.1149/1945-7111/ac201c

Bolay, L. J., Schmitt, T., Hein, S., Mendoza-Hernandez, O. S., Hosono, E., Asakura, D., Kinoshita, K., Matsuda, H., Umeda, M., Sone, Y., Latz, A., & Horstmann, B. (2022). Microstructure-resolved degradation simulation of lithium-ion batteries in space applications. *Journal of Power Sources Advances*, *14*, 100083. https://doi.org/10.1016/j.powera.2022.100083

Ciria Aylagas, R., Ganuza, C., Parra, R., Yañez, M., & Ayerbe, E. (2022). cideMOD: An Open Source Tool for Battery Cell Inhomogeneous Performance Understanding. *Journal of The Electrochemical Society*, *169*(9). https://doi.org/10.1149/1945-7111/ac91fb

Clark, S., Raynaud, X., Møll Nilsen, H., & Johansson, A. (2026). Advanced Coupled Electrochemical-Thermal Simulations of 4680 Cylindrical Cells on a Pseudo-Four-Dimensional Grid. *Journal of The Electrochemical Society*, *173*(7), 070503. https://doi.org/10.1149/1945-7111/ae4896

COMSOL AB. (2025). *COMSOL Multiphysics®* (Version 6.4). http://www.comsol.com

Demidov, D. (2020). AMGCL – A C++ library for efficient solution of large sparse linear systems. *Software Impacts*, *6*, 100037. https://doi.org/10.1016/j.simpa.2020.100037

Doyle, M., Fuller, T. F., & Newman, J. (1993). Modeling of Galvanostatic Charge and Discharge of the Lithium/Polymer/Insertion Cell. *Journal of The Electrochemical Society*, *140*(6), 1526–1533. https://doi.org/10.1149/1.2221597

Eaton, J. W., Bateman, D., Hauberg, S., & Wehbring, R. (2025). *GNU Octave*. https://octave.org

Hein, S., Danner, T., & Latz, A. (2020). An Electrochemical Model of Lithium Plating and Stripping in Lithium Ion Batteries. *ACS Applied Energy Materials*, *3*(9), 8519–8531. https://doi.org/10.1021/acsaem.0c01155

Korotkin, I., Sahu, S., O'Kane, S. E. J., Richardson, G., & Foster, J. M. (2021). DandeLiion v1: An Extremely Fast Solver for the Newman Model of Lithium-Ion Battery (Dis)charge. *Journal of The Electrochemical Society*, *168*(6), 060544. https://doi.org/10.1149/1945-7111/ac085f

Lie, K.-A. (2019). *An Introduction to Reservoir Simulation Using MATLAB/GNU Octave:*

User Guide for the MATLAB Reservoir Simulation Toolbox (MRST). Cambridge University Press. https://doi.org/10.1017/9781108591416

Planden, B., Courtier, N. E., Robinson, M., Khetarpal, A., Planella, F. B., & Howey, D. A. (2025). PyBOP: A Python package for battery model optimisation and parameterisation. *Journal of Open Source Software*, *10*(116), 7874. https://doi.org/10.21105/joss.07874

Safari, M., Morcrette, M., Teyssot, A., & Delacourt, C. (2009). Multimodal Physics-Based Aging Model for Life Prediction of Li-Ion Batteries. *Journal of The Electrochemical Society*, *156*(3), A145. https://doi.org/10.1149/1.3043429

Schmitt, C., Johansson, A., Raynaud, X., Cedeño, E. J. F., Mugisa, J., Sotta, D., Martin, A., Schaeffer, N., Debruyne, C., Reynier, Y., Clark, S., & Kopljar, D. (2026). *Comprehensive parameter and electrochemical dataset for a 1 Ah graphite/LNMO battery cell for physical modelling as a blueprint for data reporting in battery research*. arXiv. https://doi.org/10.48550/ARXIV.2601.10507

Sulzer, V., Marquis, S. G., Timms, R., Robinson, M., & Chapman, S. J. (2021). Python battery mathematical modelling (PyBaMM). *Journal of Open Research Software*, *9*(1). https://doi.org/10.5334/jors.309

The MathWorks Inc. (2025). *MATLAB*. https://www.mathworks.com

Torchio, M., Magni, L., Gopaluni, R. B., Braatz, R. D., & Raimondo, D. M. (2016). LIONSIMBA: A Matlab Framework Based on a Finite Volume Model Suitable for Li-Ion Battery Design, Simulation, and Control. *Journal of The Electrochemical Society*, *163*(7), A1192. https://doi.org/10.1149/2.0291607jes

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L. B. da, Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., … Mons, B. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, *3*(1). https://doi.org/10.1038/sdata.2016.18